

## The Culmination of Art and Math: A Golden Connection

By Harshil N.

Oftentimes, mathematics is the foundation for other fields of study. For instance, many believe mathematics to be the basis of art. As the Italian Mathematician Luca Pacioli famously said, “without mathematics there is no art.” In this article, we will aim to demonstrate several key examples of the connection between mathematics and art by introducing a very special (and golden) number. The **golden ratio**, approx. 1.618, is usually denoted using  $\phi$ . Da Vinci’s *Last Supper*, as shown in **Figure 1**, puts the golden ratio on full display with its intricate spacing. Aside from *Last Supper*, The

golden ratio can also be seen in countless Renaissance classics such as Raphael’s *The School of Athens* and Michelangelo’s *The Fall and Expulsion from the Garden of Eden*. The golden ratio can be seen in primitive architecture as well. The Great Pyramid of Giza, as seen in **Figure 2**, is estimated to have an original height of 480.94 ft and a base width of 755.68 ft, forming a height to base width ratio of approximately 0.636. With the Pythagorean Theorem, mathematicians have discovered the **Kepler Triangle** – the only right triangle whose sides are in a geometric progression (see

**Figure 3**). Using Kepler’s triangle, we can create a pyramid with base width of 2 and height of  $\phi$ . Since the square root of  $\phi$  is approx. 1.272, we can deduce that the height : base width of the pyramid is approximately 0.636. Amazingly, the height to base width ratio of the our model is the exact same as the Great Pyramid of Giza, meaning that the Great Pyramids are connected to the golden ratio! With the two examples discussed, we invite your eyes to look for any other artistic forms in the real world with mathematical precision since connections are subtle, but fascinating!

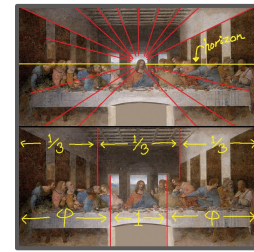


Figure 1

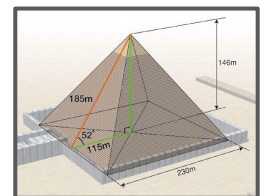


Figure 2

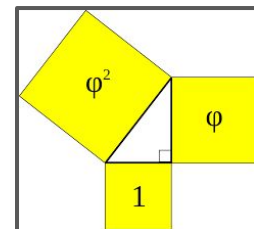


Figure 3

## Halloween Math Crossword

By Kevin Z.

Be Careful, Don’t Get Tricked...

See Answers on the Next Page!

### ONE DOWN

Little Timmy and 31 of his friends have all lost their jack-o-lanterns! Luckily, their teacher Ms. Sharpe has located them; however, she has no idea which jack-o-lantern belongs to which student. Lacking problem solving skills, Ms. Sharpe randomly returns each jack-o-lantern to a different student. If the probability that none of the children get their own jack-o-lantern back is  $x$ , find the value of  $100x$  rounded to the nearest integer.

### THREE ACROSS

Charlie Brown, being a popular guy at his high school, has been invited to attend a Halloween party this year! However, he doesn’t want to miss out on trick-or-treating at the 3 houses that offer king-sized candy bars either – Houses A (RED), B (PINK), and C (ORANGE). Luckily for Charlie, Houses A, B, and C all exist on the  $4 \times 4$  grid between his house (YELLOW) and the party. Charlie plans to determine a round-trip route which takes the shortest distance possible to hit Houses A, B, and C while also attending the party. If Charlie chooses not to pass the same house twice in fear of being recognized for candy fraud, there exists  $x$  different possible routes that he can take while staying on the lattice lines of the grid. Find the value of  $x - 404$ .

### ONE ACROSS

After trick-or-treating, Sally organizes her candies in groups of 5 and has 4 leftover; then, she eats the leftover candies. Sally then arranges the remaining candies into groups of 4 and has 2 leftover; again, she eats the leftover candies. Sally rearranges her remaining candies one last time into groups of 3 and has 1 leftover. What is the smallest number of candies that Sally could have initially had?

### TWO DOWN

Vista Way is a road in Idaho which is renowned for its decorations on Halloween. The house numbers of the 28 houses on Vista Way that have candy are the first 28 positive perfect squares. Find the smallest  $n$  such that  $n!$  is divisible by each of these house numbers.

